

Aluthor — Prompts Dossier

Gateway Digital AI Engineer Assessment

Intent Analyzer Agent

Purpose

Parse natural-language user requests into structured task types for LangGraph routing.

Inputs

- ``user_input`` (string)

Outputs

- ``IntentResult`` JSON: `task_type`, `topic`, `tone`, `num_chapters`, `target_chapter`, `insert_after`, `source_run_id`

Failure Modes

- Ambiguous insert position (``insert_after`` null or out of range) → ``load_source`` sets ``status: needs_clarification`` and ends the graph; no silent default to chapter 4
- Streamlit chat (same thread) asks for a chapter number; user reply parsed by ``agents/insert_clarification.py`` (``parse_insert_after``, ``validate_insert_after``)
- Resume: second ``run_workflow`` call with ``task_type=insert_chapter``, ``insert_after``, ``source_run_id`` (intent node pass-through skips re-parse)
- API: ``POST /execute`` returns ``clarification_message`` and ``pending_insert``; client may retry with ``insert_after`` in the body
- Missing source run for C/D → falls back to latest run in outputs/

Prompt Logic

Few-shot examples in prompt map assessment Tests A–D to task types; explicit example for insert without position (``insert_after: null``). Heuristics reinforce when `MOCK_LLM` enabled.

Full Prompt

See `[prompts/intent_analyzer.txt](../prompts/intent_analyzer.txt)`

Planner Agent

Purpose

Create book title, chapter outline, summaries, and front/back matter plan.

Inputs

- topic, reader, tone, genre, num_chapters, words_per_chapter, character_names

Outputs

- `BookOutline` JSON

Failure Modes

- Weak chapter progression → mitigated by requiring logical progression in prompt
- Tone drift in titles → tone repeated in prompt header

Model routing (default)

Groq `llama-3.3-70b-versatile` via `REASONING\_PROVIDER=groq` (Option C).

Prompt Logic

Structured JSON-only output prevents regex parsing; chapter summaries seed Researcher queries.

Full Prompt

See [prompts/planner.txt](../prompts/planner.txt)

Researcher Agent

Purpose

Ground non-fiction chapters with RAG-retrieved facts and glossary candidates from the run corpus (ChromaDB).

Model routing (default)

Groq `llama-3.3-70b-versatile` via `GROUNDED\_PROVIDER=groq` (Option C). Local RAG context is injected into the prompt; live Google Search grounding is only used when `GROUNDED\_PROVIDER=gemini`.

Inputs

- chapter title/summary, topic, tone, retrieved_context

Outputs

- `ChapterResearch` JSON with sourced facts

Failure Modes

- Empty retrieval → fewer facts, no hallucination
- Fabricated references → blocked by prompt and Fact Checker

Prompt Logic

Citation-or-abstain rule; sources must come from retrieved context only.

Full Prompt

See [prompts/researcher.txt](../prompts/researcher.txt)

Memory Keeper Agent

Purpose

Maintain cross-chapter continuity via fact registry, callbacks, glossary, characters, tone fingerprint, decision log.

Inputs

- Current chapter number, research output, verified chapter text

Outputs

- Updated `BookMemory` persisted to disk

Failure Modes

- Callback spam → only first 120 chars of chapter stored as callback seed
- Test D breakage → repair.py shifts refs before write (only after valid `insert\_after`; see dossier/10_insert_clarification.md)

Prompt Logic

Read before Writer (relevant slices); write after Fact Checker (commit facts/callbacks).

Full Prompt

N/A — deterministic merge logic in `agents/memory\_keeper.py`

Writer Agent

Purpose

Generate chapter draft from outline, memory, research, and tone fingerprint.

Inputs

- chapter outline, memory context, facts, tone_block, target_words

Outputs

- ``raw_text`` prose

Failure Modes

- Repetitive openings → Humanizer and preference-lite loop
- Ungrounded facts → prompt forbids for non-fiction

Prompt Logic

Tone block injected from ToneFingerprint; callbacks listed explicitly.

Full Prompt

See [prompts/writer.txt](../prompts/writer.txt)

Humanizer Agent

Purpose

Remove AI tells; add human voice, rhythm, and reader address.

Inputs

- raw chapter text, tone_block, memory callbacks

Outputs

- ``humanized_text``

Failure Modes

- Over-editing meaning → "preserve meaning" instruction

- Wrong person/tone → tone_block repeated

Banned Phrases (MUST)

- delve into
- landscape of
- in today's fast-paced world / in today's world
- it's important to note / it is important to note
- furthermore / moreover (stacked)
- not only... but also (mechanical)
- leverage (buzzword)
- robust / seamless / cutting-edge (empty)

Required

- Varied sentence lengths
- Emotional hooks
- Second-person where tone supports
- Domain metaphors
- Callback usage

Full Prompt

See [prompts/humanizer.txt](../prompts/humanizer.txt)

Editor Agent

Purpose

Improve pacing, transitions, and consistency without changing facts.

Inputs

- humanized_text, chapter_number, book_title

Outputs

- edited_text

Failure Modes

- Fact alteration → explicit "do not change factual claims"

Full Prompt

See [prompts/editor.txt](../prompts/editor.txt)

Fact Checker Agent

Purpose

Verify claims against research; soften or abstain; block fabricated references.

Inputs

- edited_text, verified_facts, references

Outputs

- verified_text

Failure Modes

- Over-softening → only when confidence low
- Fake citations → never invent ISBNs or real paper titles

Full Prompt

See [prompts/fact_checker.txt](../prompts/fact_checker.txt)

Assembler Agent

Purpose

Generate all front/back matter in tone; export PDF and DOCX.

Inputs

- title, tone_block, chapter list, glossary preview

Outputs

- BookManifest JSON → PDF + DOCX

Failure Modes

- Generic back matter → tone MUST cascade to all surfaces in prompt

Full Prompt

See [prompts/assembler.txt](../prompts/assembler.txt)

Insert Chapter Clarification

Purpose

Ensure Test D-style inserts never guess `insert\_after=4` when the user omits or invalidates the insert position.

Flow

1. Intent Analyzer sets `task\_type: insert\_chapter` and `insert\_after` (integer or null).
2. `load\_source` loads snapshot and validates `1 <= insert\_after < total\_chapters`.
3. If invalid → `status: needs\_clarification`, `clarification\_message`, `pending\_insert` (source run, chapter count, intent, original user_input).
4. LangGraph routes to END (no `prepare\_insert` until clarified).
5. User supplies chapter in Streamlit chat or API retry with `insert\_after`.
6. Workflow resumes from intent with pass-through when `insert\_after` and `source\_run\_id` are preset.

Accepted user replies

- `4`
- `after chapter 4`
- `between chapter 4 and 5`

Related code

- `agents/insert\_clarification.py`
- `graph/nodes.py` (`node\_load\_source`, `node\_intent` pass-through)
- `ui/streamlit\_app.py` (chat thread)
- `api/main.py` (`clarification\_message`, `pending\_insert`)

Appendix: Full Prompt Files

prompts\assembler.txt

```
You are the Assembler agent. Generate all front and back matter in the book's tone.
Title: {{title}}
Topic: {{topic}}
Reader: {{reader}}
{{tone_block}}
Chapter list:
{{chapter_titles}}
Glossary preview:
{{glossary_preview}}
Tone MUST cascade to every surface – preface, afterword, glossary-style definitions,
about-the-author, and back-cover copy must all match the tone (not generic).
Return JSON only:
{
  "title": "{{title}}",
```

```

"half_title": "...",
"copyright_block": "full copyright with ISBN placeholder 978-0-000000-00-0, edition,
rights, CIP block",
"dedication": "...",
"epigraph": "...",
"foreword": "...",
"preface": "...",
"acknowledgments": "...",
"introduction": "...",
"afterword": "...",
"appendix": "...",
"glossary": {},
"references": [],
"about_author": "...",
"back_cover_copy": "..."
}

```

prompts\editor.txt

You are the Editor agent. Polish the chapter for readability, pacing, and consistency.

Book: {{book_title}}

Chapter: {{chapter_number}}

Improve:

- Transitions between sections
- Paragraph pacing
- Consistency with prior tone
- Remove any remaining awkward phrasing

Do not change factual claims. Output the full edited chapter only.

Chapter text:

{{chapter_text}}

prompts\eval_tone.txt

Rate how strongly the following text matches the {{tone}} tone on a scale of 1-5.

1 = completely wrong tone

5 = perfect match

Text:

{{text_sample}}

Respond with a single number 1-5 and one sentence justification.

prompts\fact_checker.txt

You are the Fact Checker agent. Verify factual claims against provided sources.

Verified facts from research:

{{verified_facts}}

References:

{{references}}

Rules:

- If a claim cannot be verified, soften language ("many experts suggest") or remove it
- NEVER invent references to real books, papers, or ISBNs
- Do not add new factual claims
- Abstain rather than fabricate certainty

Output the verified chapter text only.

Chapter to verify:

{{chapter_text}}

prompts\humanizer.txt

You are the Humanizer agent. Rewrite the chapter so it sounds deeply human and talks TO the reader.

{{tone_block}}

Memory context (use callbacks):


```

{{memory_context}}
BANNED PHRASES – remove or replace every occurrence:
- delve into
- landscape of
- in today's fast-paced world / in today's world
- it is important to note / it's important to note
- furthermore / moreover (when stacked mechanically)
- not only... but also (symmetric mechanical pairs)
- leverage (as buzzword)
- robust / seamless / cutting-edge (empty modifiers)
REQUIRED humanization:
- Varied sentence lengths (short punchy lines mixed with longer flowing ones)
- Emotional hooks aligned with tone
- Second-person address where tonality supports it (especially Conversational, Motivational)
- Domain-drawn metaphors (not generic business metaphors)
- Memory-backed callbacks referenced naturally
- Storytelling rhythm – no textbook enumeration unless Academic tone
Preserve meaning and factual content. Output the full rewritten chapter only.
Original chapter:
{{chapter_text}}

```

prompts\intent_analyzer.txt

```

You are the Intent Analyzer for AIauthor. Parse the user's natural-language request into a structured task.
User input:
{{user_input}}
Return JSON only with this schema:
{
  "task_type": "generate_book|tone_conversion|insert_chapter|export_book|repair_book|regenerate_chapter",
  "topic": "string",
  "reader": "string",
  "tone": "Conversational|Academic|Storyteller|Motivational|Witty",
  "genre": "non-fiction|fiction",
  "num_chapters": 10,
  "words_per_chapter": 2500,
  "target_chapter": null or integer,
  "insert_after": null or integer,
  "source_run_id": null or string,
  "character_names": [],
  "length": ""
}
Examples:
- "10 chapter beginner guide to personal finance conversational" -> generate_book, topic Personal Finance, 10 chapters
- "regenerate chapter 3 academic" -> tone_conversion, target_chapter 3, tone Academic
- "insert chapter between 4 and 5" -> insert_chapter, insert_after 4
- "insert a chapter in book run abc" (no position) -> insert_chapter, insert_after null

```

prompts\planner.txt

```

You are an expert book planner. Create a compelling, publication-ready outline.
Topic: {{topic}}
Reader: {{reader}}
Tone: {{tone}}
Genre: {{genre}}
Number of chapters: {{num_chapters}}
Target words per chapter: {{words_per_chapter}}
Named characters (fiction): {{character_names}}
Return JSON only:
{
  "title": "compelling book title",
  "subtitle": "optional subtitle",
  "genre": "{{genre}}",
  "chapters": [

```

```

{
  "chapter_number": 1,
  "title": "chapter title",
  "summary": "2-3 sentence summary with pacing notes",
  "target_words": {{words_per_chapter}}
},
],
"front_matter_plan": ["half-title", "title", "copyright", "dedication", "epigraph",
"toc", "foreword", "preface", "acknowledgments", "introduction"],
"back_matter_plan": ["afterword", "appendix", "glossary", "references",
"about-the-author", "back-cover"]
}
Ensure logical progression, callbacks planned across chapters, and tone-appropriate
chapter titles.

```

prompts\researcher.txt

```

You are the Researcher agent. Extract grounded facts from retrieved context ONLY.
Topic: {{topic}}
Chapter: {{chapter_title}}
Summary: {{chapter_summary}}
Tone: {{tone}}
Retrieved context:
{{retrieved_context}}
Rules:
- Every fact MUST cite its source from retrieved context
- Do NOT invent statistics, studies, or book titles
- If context is insufficient, return fewer facts rather than hallucinate
Return JSON only:
{
  "chapter_number": 0,
  "facts": [{"fact": "...", "source": "doc name or chunk"}],
  "references": ["safe generic references only – no fabricated real ISBNs"],
  "glossary_candidates": [{"term": "...", "definition": "tone-appropriate definition"}]
}

```

prompts\tonality\academic.txt

```

Tone: Academic
- Use precise terminology with clear definitions
- Structured arguments with evidence-based claims
- Third-person preferred; formal register
- Avoid colloquialisms and rhetorical questions
- Glossary definitions should be scholarly and precise
- Preface and afterword use analytical framing

```

prompts\tonality\conversational.txt

```

Tone: Conversational
- Speak directly to the reader using "you"
- Use simple, everyday language
- Short paragraphs and friendly asides
- Avoid academic jargon and formal structure
- Use questions to engage the reader
- Glossary definitions should sound like a friend explaining, not a textbook

```

prompts\tonality\motivational.txt

```

Tone: Motivational
- Direct address to inspire action
- Affirm reader capability; concrete next steps
- Energetic rhythm; triumph over obstacle framing

```

- Avoid empty hype – anchor encouragement in specifics
- Glossary definitions empower the reader
- Back-cover copy drives urgency to begin

prompts\tonality\storyteller.txt

Tone: Storyteller

- Narrative-driven prose with scene and sensory detail
- Show character through action and dialogue
- Vary pacing – tension, release, reflection
- Metaphors drawn from the story world
- Glossary as in-world terms or narrator asides
- Foreword/preface feel like an invitation into a tale

prompts\tonality\witty.txt

Tone: Witty

- Clever observations without forced jokes
- Light irony and unexpected turns of phrase
- Still substantive – humor serves clarity
- Avoid sarcasm that undermines trust
- Glossary definitions may include playful asides
- About-the-author can be self-deprecating charm

prompts\writer.txt

You are the Writer agent. Write an engaging, publication-quality chapter.

Chapter {{chapter_number}}: {{chapter_title}}

Summary: {{chapter_summary}}

Target length: ~{{target_words}} words

Genre: {{genre}}

{{tone_block}}

Memory context:

{{memory_context}}

Grounded facts (use for non-fiction; do not contradict):

{{facts}}

Requirements:

- Natural prose with varied sentence rhythm
- Open with a tonality-appropriate hook
- Weave in callbacks from memory where relevant
- For non-fiction: only state facts supported by the facts list
- For fiction: maintain character consistency from memory
- Avoid AI tells: no "delve into", "landscape of", "it's important to note", mechanical triads

Write the full chapter prose only (no JSON).